

Figure 13.1 The padlock provides visible confirmation that the connection is secure.

When you visit a site over an unsecured connection, most browsers provide a warning instead of a padlock. The warning consists of a triangular sign and the words *Not secure*, which can be seen in figure 13.2.



Figure 13.2 The warning triangle and words Not secure confirm this is an unsecured connection.

When you deploy your website to a public web server, you will obtain your TLS certificate from a trusted issuer, so that browsers accept it. How you go about this process depends on the issuer that you use. You should check their documentation or support. When developing and testing your application locally, you also need a certificate the browser trusts to run the application over HTTPS.

13.1.2 Using HTTPS in development

When you install the .NET 6 SDK, it installs a TLS certificate for development purposes (the ASP.NET Core HTTPS development certificate), which is used by the Kestrel server. By default, this certificate is not trusted. When you first run any application using Visual Studio on its configured HTTPS port, you are prompted to trust the certificate. If you are not a Visual Studio user, you can use the `dotnet` tool to trust the certificate, using the following command:

```
dotnet dev-certs https --trust
```